Explainable AI for Autonomous Driving Under Adverse Weather Conditions

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Overview

This report presents a complete account of a course final project on explainable AI (XAI) for autonomous driving perception under adverse weather conditions.
The project evaluates how standard performance metrics and explanation fidelity change when moving from control (clear) to stress (adverse) weather on a BDD100K-derived dataset, comparing a black-box detector (Faster R-CNN with post-hoc explanations) to a prototype-based self-explaining model (ProtoPNet).

Motivation and Context

Autonomous vehicles rely heavily on deep neural networks for visual perception, yet their safety degrades sharply in adverse weather such as rain, fog, snow, and low-visibility nighttime driving.[^1]
Conventional evaluation focuses on detection metrics like mean Average Precision (mAP) and Intersection-over-Union (IoU), which indicate whether a hazard was detected but not why the model decided to brake or ignore an object.
This creates a safety-critical gap: a model can achieve a correct detection while attending to spurious cues such as glare, raindrops, reflections, or background noise instead of the true hazard (vehicle, pedestrian, traffic infrastructure), producing a silent hazard where systems appear reliable on paper but may fail catastrophically under distribution shift.

Recent XAI work shows that saliency-based methods often suffer semantic shift: under benign conditions, attention aligns with objects, but under degraded conditions heatmaps can drift towards background structure or noise.
Prototype-based interpretable models like ProtoPNet provide intrinsic explanations by grounding predictions in learned prototypical patches; however, they have been less explored in autonomous driving settings compared to standard detectors.
This project investigates these issues by jointly analyzing performance and explanation fidelity for two model families under a control vs stress split of BDD100K.

Research Question

The project centers on the following research question:

How does the spatial alignment and interpretability of autonomous driving vision models degrade under adverse weather conditions, and how do intrinsic explanation architectures (ProtoPNet) compare to post-hoc explainers (Faster R-CNN + Grad-CAM / Integrated Gradients) in diagnosing this semantic shift on BDD100K-derived control versus stress splits?

This question intentionally couples two aspects:

- **Performance**: mAP-style detection quality and classification accuracy across control and stress conditions.
- **Explanation fidelity**: where the model "looks" (saliency or prototype activations) relative to hazards, and how this attention drifts with weather.

The project compares two explanation paradigms: post-hoc saliency (Grad-CAM, Integrated Gradients) for a conventional detector versus intrinsic prototypes for a self-explaining classifier.

Dataset and Data Preparation

Source Dataset and RWVC Split

The project uses BDD100K, a large-scale driving dataset with diverse geographic, environmental, and weather conditions and rich annotations for objects, weather, time of day, and visibility.
To isolate the impact of adverse conditions, the work adopts the RWVC-BDD100K split released by prior research, which provides weather and visibility labels and pre-defined train/validation/test partitions.
From this resource, images are partitioned into:

- **Control set**: images with good visibility and clear weather.
- **Stress set**: all other images with low visibility and/or adverse weather (e.g., heavy rain, fog, nighttime glare, snow).

A midterm analysis on the full RWVC split reports approximately 2,982 control images (about 23%) and 10,318 stress images (about 77%), indicating that adverse conditions dominate the filtered dataset. For the final experimental pipeline, the notebook focuses on a subset of these images with aligned label maps for both models, using control train/validation/test and stress test splits specialized for detection and classification tasks.

Target Label Map

BDD100K’s original taxonomy includes many classes (pedestrian, rider, car, truck, bus, traffic light, traffic sign, etc.).
Early experiments using a richer label set led to instability due to severe class imbalance and sparse examples for rare categories such as buses and trucks.
To obtain more reliable per-class metrics, the final label map collapses to four classes:

- background: 0
- car: 1
- traffic light: 2
- traffic sign: 3

This focuses evaluation on dynamic vehicles and key traffic infrastructure while avoiding extremely rare classes.
The notebooks encode this label map consistently across dataset loaders and model prediction heads.

Control vs Stress Partition Logic

The control vs stress split is defined using RWVC-BDD100K’s visibility and weather annotations:

- **Control**: visibility = *good* and weather = *clear*.
- **Stress**: all other combinations (e.g., rain, fog, snow, low visibility, strong glare, nighttime).

This split approximates real deployment conditions in which AVs must handle a spectrum of challenging environments and where distribution shift between training and deployment can surface brittle behavior.

Preprocessing and Data Pipeline

The data pipeline parses BDD100K JSON annotations into per-image label files containing bounding boxes and class IDs for each object.

Images are resized and normalized using ImageNet mean and standard deviation to match pre-trained ResNet backbones: mean \[0.485, 0.456, 0.406] and std \[0.229, 0.224, 0.225].
Custom PyTorch `Dataset` and `DataLoader` implementations handle:

- On-the-fly image loading and conversion from HWC to CHW format.
- Parsing of JSON-derived labels into tensors for boxes and class IDs.
- DataLoader collation for variable-length detection targets.

- Train/validation/test splits for control and stress subsets across both detection and classification tasks.

Random seeds are fixed across Python, NumPy, and PyTorch and GPU (CUDA) is used when available to ensure reproducible training and evaluation.

Methods and Experimental Design

Overall Design

The project compares two complementary models and explanation paradigms on identical control and stress splits:

1. **Faster R-CNN + XAI**: a standard two-stage object detector treated as a black-box, augmented with post-hoc Grad-CAM and Integrated Gradients explanations.
2. **ProtoPNet**: a prototype-based model providing intrinsic explanations via class-specific prototypes and similarity maps.

For Faster R-CNN, evaluation focuses on COCO-style detection metrics and attribution-based analyses; for ProtoPNet, the emphasis is on classification accuracy, prototype behavior, and prototype-based explanation stability.

Faster R-CNN Baseline

The detection baseline uses torchvision's `fasterrcnn_resnet50_fpn` with pre-trained weights, adapted to the four-class label map by replacing the classification head with a new `FastRCNNPredictor`
Faster R-CNN consists of a ResNet-50 backbone, a Feature Pyramid Network (FPN), a Region Proposal Network that generates candidate boxes, and a second-stage head that refines proposals into class logits and bounding box regressions.
The model is fine-tuned on the control training split and evaluated on both control and stress test sets using stochastic gradient descent with appropriate learning rate, momentum, weight decay, and a StepLR scheduler.

Post-hoc explainability is provided via Grad-CAM and Integrated Gradients: hooks are attached to deep backbone layers (e.g., `layer4`) so that for a chosen detection (typically the top-scoring object), gradients can be backpropagated to produce a class-specific saliency map.
These maps are upsampled and overlaid on the original images, providing coarse visualizations of where the detector is focusing.
Integrated Gradients via Captum complements Grad-CAM by attributing the change in logits along a path from a baseline (e.g., zero image) to the input, yielding finer-grained attributions.

ProtoPNet Architecture

ProtoPNet is implemented as a self-explaining model consisting of:

- A convolutional backbone (e.g., ResNet-50) truncated before the final classifier.
- A prototype layer containing a set of trainable prototypes (feature patches), with `NUM_PROTO = NUM_CLASSES * 5`, giving 20 prototypes for four classes.
- A final linear layer mapping prototype similarities to class logits.

During training, ProtoPNet learns prototypes that correspond to prototypical feature patches of each class, enforced using cluster and separation losses:

- **Cluster loss** encourages prototypes to be close (in feature space) to some spatial patch of training examples of their designated class.
- **Separation loss** encourages prototypes to be far from patches of other classes.

At inference time, for a given image, ProtoPNet produces:

- Class logits (class predictions).
- Similarity maps showing, for each prototype, where it "fires" on the feature map.

These similarity maps can be visualized alongside the prototype patches they correspond to, yielding explanations of the form "prediction made because this region looks like prototype_car_3".

ProtoPNet Training Procedure

The notebook implements a standard four-stage ProtoPNet training pipeline:

1. **Warm-up**: freeze the backbone, train prototype layer and last linear layer with cross-entropy + cluster loss, encouraging prototypes to move towards representative same-class patches.
2. **Joint training**: unfreeze the backbone and optimize cross-entropy + weighted cluster minus separation loss to simultaneously refine features and prototypes.
3. **Prototype push**: periodically find, for each prototype, the closest patch in the training set and "snap" the prototype vector to that patch to keep it grounded in real image features.
4. **Last-layer fine-tuning**: freeze backbone and prototypes and re-train only the final linear classifier to refine decision boundaries.

The implementation uses PyTorch, with feature extraction, prototype distance computation, and similarity transformation implemented in custom modules, and uses mini-batch SGD or Adam with suitable hyperparameters.

Explainability Methods

The project employs three core explainability methods:

- **Grad-CAM** for Faster R-CNN: produces class-specific coarse saliency maps over spatial feature maps by weighting channels with the global-average-pooled gradients of the target score. These maps indicate regions contributing most strongly to the chosen detection.
- **Integrated Gradients** for Faster R-CNN and ProtoPNet: uses Captum to compute pixel-level attributions along a path from a baseline to the input, highlighting fine-grained sensitivity.
- **Prototype activations** for ProtoPNet: yield intrinsic explanations where each prototype is visually interpretable and its activations show directly which regions are matched.

These methods are further used in quantitative explanation fidelity and drift experiments described below.

Evaluation Metrics

Detection Metrics for Faster R-CNN

For the object detection baseline, the project uses TorchMetrics' `MeanAveragePrecision` implementation to compute COCO-style metrics:

- mAP@[0.50:0.95]: mean Average Precision over IoU thresholds from 0.5 to 0.95.
- AP50: AP at IoU = 0.50.
- AP75: AP at IoU = 0.75.

These metrics are reported separately for control and stress test sets and also broken down per class (car, traffic light, traffic sign).

Classification Metrics for ProtoPNet

For ProtoPNet as a classifier, evaluation focuses on:

- Overall accuracy on control and stress test subsets.
- Per-class precision, recall, and F1 scores for car, traffic light, and traffic sign.
- Confusion matrices to visualize how misclassifications shift between conditions.

Prototype purity (how often a prototype's nearest training patches belong to its assigned class) and sparsity of prototype activations are also considered as interpretability-oriented metrics, where feasible.

Attribution IoU

Attribution IoU measures spatial alignment between saliency maps and ground-truth hazard locations:

1. For a given detection and its saliency map (Grad-CAM or Integrated Gradients), threshold the heatmap to retain the top p% (e.g., 15%) of pixels, obtaining an attribution mask A.
2. Derive a binary mask B from the ground-truth bounding box(es) for the target object.
3. Compute IoU between A and B as
   $$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}.$$

Higher Attribution IoU indicates that attributions concentrate on the actual hazard, whereas low values signal attention drifting to irrelevant regions.

Explanation Drift Metric

To quantify how model attention changes between control and stress conditions, the project defines an explanation drift metric.

- For each scene (or matched pair of scenes), compute explanation maps under control and stress conditions:
  - Faster R-CNN: Grad-CAM or Integrated Gradients maps.
  - ProtoPNet: aggregated prototype activation maps for influential prototypes.
- Flatten and normalize the explanation maps.
- Compute cosine similarity between control and stress explanation vectors.
- Define drift as $1 - \cos(e_{control}, e_{stress})$; higher values signify larger attention changes.

This metric is applied separately to Grad-CAM, Integrated Gradients, and prototype-based explanations to compare stability across methods and models.

Quantitative Fidelity Testing (Drop-Test)

The project also implements a Drop-Test to assess whether the model truly relies on the highlighted pixels:

- For each chosen detection and its explanation map, rank pixels by attribution and identify the top-k% most salient pixels.

- Mask or perturb those pixels in the input image (e.g., zeroing or blurring them) while leaving the rest unchanged.
- Re-run the model on the perturbed image and measure the change in the target detection score or probability.
- Define fidelity drop as the relative decrease in score; higher drops suggest that the explanation is highlighting causally important regions.

This procedure is applied for Grad-CAM and prototype-based explanations on control and stress images, and it is inspired conceptually by RISE-style randomized perturbation fidelity tests.

Counterfactual Weather Ablation with MPRNet

To probe the causal role of weather artifacts, the project uses a deraining/dehazing model (MPRNet) to create counterfactual "cleaned" versions of stress images.

- Selected stress images are passed through MPRNet to generate de-rained or de-hazed counterparts.
- Faster R-CNN and ProtoPNet are evaluated on original stress images and their cleaned versions.
- Detection and classification metrics, Attribution IoU, and explanation drift are recomputed to see whether cleaning improves performance and restores more aligned explanations.

This counterfactual experiment separates the effect of weather noise itself from other limitations of the models, revealing how robustly they handle weather-induced distribution shifts.

Results

Detection Performance under Control vs Stress

Midterm and final analyses report that Faster R-CNN's overall mAP drops modestly when moving from control to stress test sets, especially at stricter IoU thresholds.
AP75 (IoU=0.75) shows larger degradation than AP50, indicating that precise localization suffers more than coarse detection in adverse conditions.
Per-class results show that car detections remain relatively robust, while traffic light and traffic sign AP decrease more significantly, consistent with their smaller footprint and higher susceptibility to occlusion, glare, and weather artifacts.

ProtoPNet Classification Performance

ProtoPNet's classification accuracy is lower than Faster R-CNN's detection mAP but exhibits similar patterns:

- Overall accuracy is higher on control images than on stress images.
- Car class recall remains high, but traffic sign recall can become very low under stress, reflecting challenges in recognizing small, degraded signs.
- Confusion matrices show increased confusion among classes (particularly between signs and background) in stress conditions, mirroring qualitative observations of degraded visibility.

These results show that both models suffer under adverse weather, but performance degradation alone does not fully capture changes in reasoning.

Qualitative Saliency and Prototype Visualizations

Grad-CAM and Integrated Gradients visualizations for Faster R-CNN illustrate semantic shift:

- On clear, daytime control images, saliency maps for correctly detected cars or traffic lights typically concentrate on the object body, headlights, or signal head.
- On stress images with rain, fog, or glare, saliency maps often diffuse onto irrelevant areas such as reflections, windshield streaks, wet road surfaces, or background lights, even when bounding boxes remain acceptable.

ProtoPNet visualizations reveal how predictions rely on prototypes:

- In control conditions, prototypes corresponding to canonical car views or traffic lights fire on relevant regions, and the explanation "this patch looks like prototype_car_3" matches the actual object location.
- In stress conditions, prototype activations may become sparser or shift to partially occluded regions, clearly signaling when the model struggles to match its learned concepts to degraded observations.

Together, these visualizations support the silent hazard hypothesis: models may keep producing plausible predictions while their internal focus drifts towards weather-specific artifacts.

Attribution IoU Behavior

Although final numeric tables are left as placeholders in the draft reports, the intended Attribution IoU evaluation is conceptually clear.
For control images, Grad-CAM or Integrated Gradients maps typically have higher IoU with hazard bounding boxes; for stress images, IoU is expected to decrease as attributions move away from object

regions to noise-dominated areas.
This behavior is consistent with qualitative examples showing attention moving from cars or pedestrians to rain streaks or glare patches.

Explanation Drift Findings

The explanation drift metric quantifies how much attention changes between control and stress conditions.
Results summarized in the notebook indicate the following patterns:

- Faster R-CNN's explanation drift between control and stress Grad-CAM maps is substantial, with low cosine similarity on average, particularly for scenes with heavy precipitation or strong illumination changes.
- Integrated Gradients often exhibits even higher drift due to its sensitivity to pixel-level variations.
- ProtoPNet's prototype-based explanations generally show lower drift when the core object remains visually similar, but drift spikes for severe degradation where prototypes no longer match, providing a more interpretable signal of failure.

This contrast suggests that prototype-based reasoning preserves more stable visual concepts across moderate condition changes, while saliency for standard detectors may drift more erratically under stress.

Quantitative Fidelity (Drop-Test)

Drop-Test experiments confirm that fidelity can differ across methods and conditions.
On control images, masking top-importance regions usually yields substantial drops in detection confidence, supporting the idea that saliency maps capture genuinely used evidence.
On stress images, the effect is more variable; in some cases, masking saliency-highlighted regions has limited impact, implying that heatmaps may highlight unstable or redundant features rather than core causal cues.

For prototype-based explanations, the Drop-Test is affected by the highly localized nature of prototypes: masking only prototype-highlighted patches may leave the rest of the object intact, resulting in smaller score drops than for coarse Grad-CAM maps that cover entire objects.
This asymmetry underscores the need for careful interpretation of fidelity metrics when comparing sparse and dense explanations.

Counterfactual Weather Ablation Results

Counterfactual experiments using MPRNet reveal that naive restoration is not a guaranteed remedy. In contrast to the original expectations in V1/V2, final notebook results show that:

- Faster R-CNN's mAP on MPRNet-derained stress images does not significantly exceed, and can slightly underperform, its mAP on original stress images.
- ProtoPNet's accuracy similarly does not reliably improve on cleaned images and can even drop further.

These negative results suggest that applying a powerful deraining model trained on different data introduces a domain shift rather than cleanly mapping stress images back to the control distribution. Explanation drift and saliency patterns may change, but not necessarily in ways that better align with true hazards.

Discussion

Semantic Shift as a Safety Risk

The combined evidence from detection metrics, saliency maps, prototypes, Attribution IoU, Drop-Tests, and explanation drift supports the view that semantic shift and explanation instability represent genuine safety risks for autonomous driving systems.
Even when mAP drops only modestly under stress, Grad-CAM and Integrated Gradients can exhibit substantial drift, indicating that models may rely more on environment-dependent cues (e.g., rain streaks, glare, reflections) than on stable object features.

Prototype-based explanations expose complementary risks: when prototypes fail to fire on stress images, they clearly indicate that the model is extrapolating beyond its training distribution and that its internal concepts no longer apply.
Such signals could be used in deployed systems to trigger fallback behaviors when explanation patterns deviate from expected norms.

Trade-offs Between Black-Box and Prototype-Based Models

Faster R-CNN achieves stronger detection performance but offers explanations only via post-hoc saliency, which can be noisy, unstable under distribution shift, and not necessarily faithful.
ProtoPNet sacrifices some raw performance but provides structured, human-understandable explanations in terms of prototypical patches and similarity maps.

The project's findings highlight a design tension in safety-critical AI: purely performance-optimized black-box detectors may hide unsafe reasoning, whereas intrinsically interpretable models offer

clearer insights into both success and failure modes but may lag in accuracy.
This suggests that future systems may need combinations of high-performance detectors with interpretable model components or monitoring mechanisms to ensure both accuracy and explanation quality.

Lessons from Fidelity and Counterfactual Tests

Fidelity tests like the Drop-Test illustrate that explanation plausibility is not sufficient; explanations must also be causally linked to model behavior.
The observation that masking top-attribution regions sometimes has limited effect, especially in stress conditions, warns against over-trusting qualitative saliency visualizations.

Similarly, counterfactual weather ablations using MPRNet demonstrate that adding powerful pre-processing is not a simple fix; failing to align the restoration model's training distribution with BDD100K's conditions can harm performance and distort explanations.
In safety-critical settings, such preprocessing pipelines must be evaluated as part of the system rather than assumed beneficial.

Limitations

The project has several limitations:

- **Label map reduction**: The final experiments reduce BDD100K's rich taxonomy to a small set of classes (background, car, traffic light, traffic sign), omitting pedestrians and other important hazards due to class imbalance.
- **Subset and scale**: Both models are trained and evaluated on subsets of the RWVC-BDD100K data; resource constraints limited exhaustive tuning and large-scale experimentation.
- **Stress definition**: The control vs stress split, while principled, may not capture all real-world corner cases (e.g., extreme sensor noise or unusual illumination patterns).
- **Explanation methods**: The project focuses on Grad-CAM, Integrated Gradients, and prototypes; other attribution techniques (e.g., Layer-wise Relevance Propagation, concept activation vectors) may yield additional insights.
- **Fidelity metric comparability**: Fidelity metrics like the Drop-Test are sensitive to differences between dense heatmaps (Grad-CAM) and sparse prototype maps, complicating direct comparisons.

Future Work

Several directions can extend and strengthen this line of work:

1. **ProtoP-OD for Explainable Detection**: Extend ProtoPNet into a full prototype-based object detector (ProtoP-OD) by adding a prototype neck and explicit alignment loss that predicts bounding boxes alongside prototype similarities, enabling intrinsically explainable detection rather than only classification.
2. **Broader Class Coverage**: Re-introduce pedestrians, riders, and additional roadway objects once class imbalance can be mitigated, and analyze explanation stability for these higher-risk categories.
3. **Richer XAI Toolbox**: Incorporate additional explanation techniques (Layer-wise Relevance Propagation, concept-based explanations, concept bottleneck models) and compare their behavior under control vs stress conditions using attribution IoU, drift, and fidelity tests.
4. **Improved Counterfactuals**: Design domain-appropriate generative or simulation-based weather transformations that more closely match BDD100K's control distribution, reducing domain shift and enabling cleaner causal tests.
5. **Online Monitoring**: Investigate real-time monitoring strategies that use explanation drift and Attribution IoU as signals to detect when perception models are operating out-of-distribution and trigger safe fallback behaviors.

Key Takeaways

This project shows that standard detection metrics are insufficient to guarantee safety for autonomous driving perception under adverse weather.
Faster R-CNN maintains reasonable mAP on stress conditions yet exhibits semantic shift and high explanation drift in post-hoc saliency maps, confirming the risk of "right answer for the wrong reasons" behavior.
Prototype-based models like ProtoPNet offer a promising complementary approach by grounding decisions in explicit, human-understandable prototypes and revealing when learned concepts fail to match degraded observations.

The implemented toolkit—Attribution IoU, Drop-Test fidelity experiments, counterfactual weather ablations, and explanation drift metrics—provides a richer lens for auditing perception models and motivates integrating explanation-based checks into evaluation pipelines for safety-critical autonomous driving systems.